

Project Approach (Roadmap)

Title: Business Intelligence Dashboards for Organizational Performance Analysis

Domain: Business Intelligence and Business Analytics

Dataset:

1. Superstore Sales Dataset (*sales, profit, customer, product, and regional business data*)

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

2. Global Superstore Dataset (*large-scale retail sales, orders, shipping, customer segments, and market performance data*)

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

3. Sample Financial Dataset (Microsoft Power BI Financial Sample) (*revenue, profit, discounts, units sold, product, and country-wise financial performance data*)

<https://learn.microsoft.com/en-us/power-bi/create-reports/sample-financial-download>

ABSTRACT:

Business Intelligence (BI) plays a significant role in helping organizations transform raw business data into meaningful insights for effective decision-making. The objective of this project, "**Business Intelligence Dashboards for Organizational Performance Analysis**," is to analyze organizational data and develop interactive dashboards that provide a comprehensive view of business performance.

The project utilizes benchmark business datasets containing information related to sales, customers, products, revenue, profit, and regional performance. The collected data undergoes preprocessing techniques such as data cleaning, handling missing values, removing duplicate records, and data transformation to improve its quality and consistency. Exploratory Data Analysis (EDA) is then performed to identify trends, patterns, correlations, and key business insights.

Based on the analysis, Key Performance Indicators (KPIs) such as Total Sales, Total Profit, Revenue, Customer Growth, and Regional Performance are identified and visualized using Business Intelligence tools like Microsoft Power BI. Interactive dashboards consisting of KPI cards, bar charts, line charts, pie charts, maps, and filters enable users to monitor organizational performance efficiently and make data-driven decisions.

The developed dashboard provides valuable insights into business operations, highlights high-performing and low-performing areas, and supports strategic planning through comparative analysis. Overall, the project demonstrates how Business Intelligence dashboards can enhance organizational performance monitoring and improve decision-making by presenting complex business data in a simple, interactive, and visually informative manner.

EDA Flow Diagram:

EDA Flow Diagram:

Raw Business Dataset

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Dataset Overview

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Data Quality Assessment

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Descriptive Statistics

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Missing Values Analysis

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Duplicate Records

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Outlier Detection

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Univariate Analysis



Bivariate Analysis



Multivariate Analysis



Correlation Analysis



KPI Identification



Feature Engineering (if required)



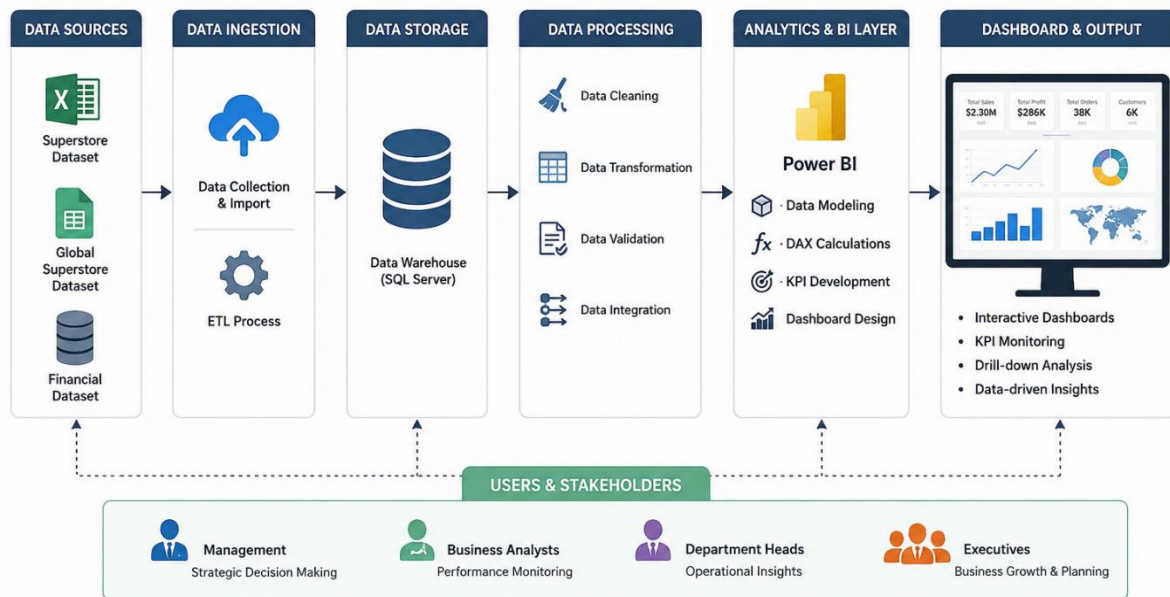
Dashboard Development



Business Performance Analysis

Architecture Diagram:

Business Intelligence Dashboard Architecture for Organizational Performance Analysis



PREPROCESSING:

- Missing Values Handling
- Duplicate Record Removal
- Data Cleaning
- Data Transformation
- Data Integration
- Data Formatting
- Data Normalization (if required)

BI Dashboard Development:

- Microsoft Power BI
- Interactive Dashboards
- KPI Cards
- Charts and Graphs
- Filters & Slicers
- Drill-down Reports

Business Insights:

- Key Performance Indicators (KPIs)
- Sales Performance Analysis
- Profitability Analysis
- Customer Behavior Analysis
- Regional Performance Analysis
- Trend Analysis

Visualization:

- KPI Cards
- Bar Charts
- Line Charts
- Pie Charts
- Treemaps
- Heatmaps
- Maps
- Tables & Matrix Reports
- Interactive Dashboard

Research Pipeline:

Business Dataset

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Data Collection

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Data Cleaning

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Data Preprocessing

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Exploratory Data Analysis (EDA)

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KPI Identification

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Dashboard Development

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Visualization

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Business Performance Analysis

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Comparative Analysis

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Business Insights & Recommendations

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Project Conclusion

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